

TechBiz Academy

Week 2 - Task 6

Vulnerable Lab Setup_Metasploitable / DVWA

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Course

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Organization

TechBizz Security

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Objective

The objective of this lab is to prepare a vulnerable penetration testing environment by installing **Metasploitable 2** and **DVWA (Damn Vulnerable Web Application)**. Both virtual machines are configured on the same virtual network so Kali Linux can communicate with the vulnerable target for future penetration testing exercises.

Introduction

A penetration testing lab requires a vulnerable target machine and an attacker machine.

In this lab:

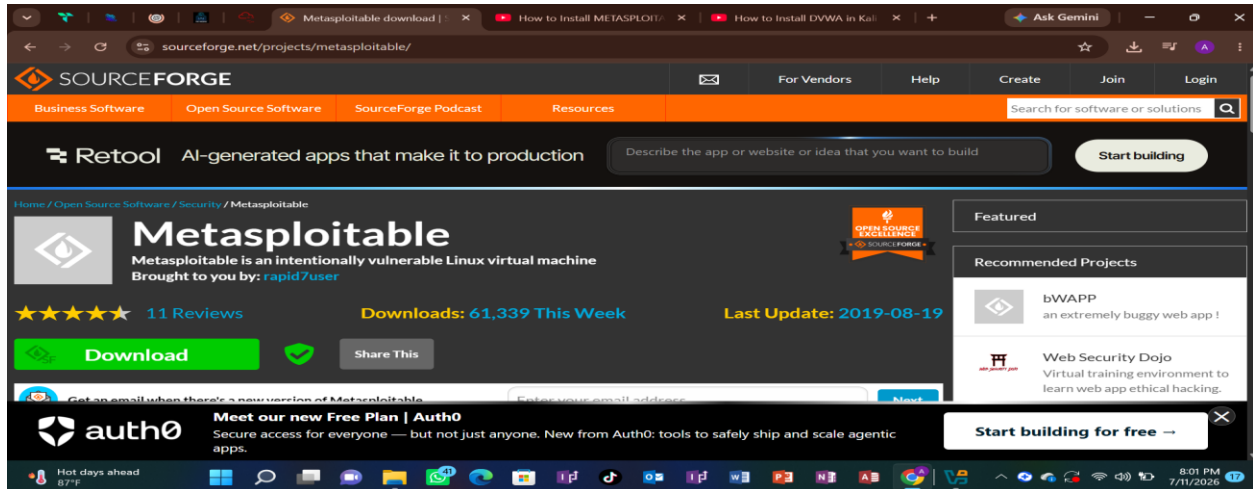
- Kali Linux acts as the attacker.
- Metasploitable 2 acts as the vulnerable target.
- DVWA provides a vulnerable web application for practicing web attacks.

Both systems are configured inside Oracle VirtualBox and connected through the same virtual network.

1. Downloading Metasploitable 2

The first step was downloading the Metasploitable 2 virtual machine from SourceForge.

The compressed archive was downloaded and extracted on the Windows host system.



(Screenshot 1: SourceForge Metasploitable Download Page)

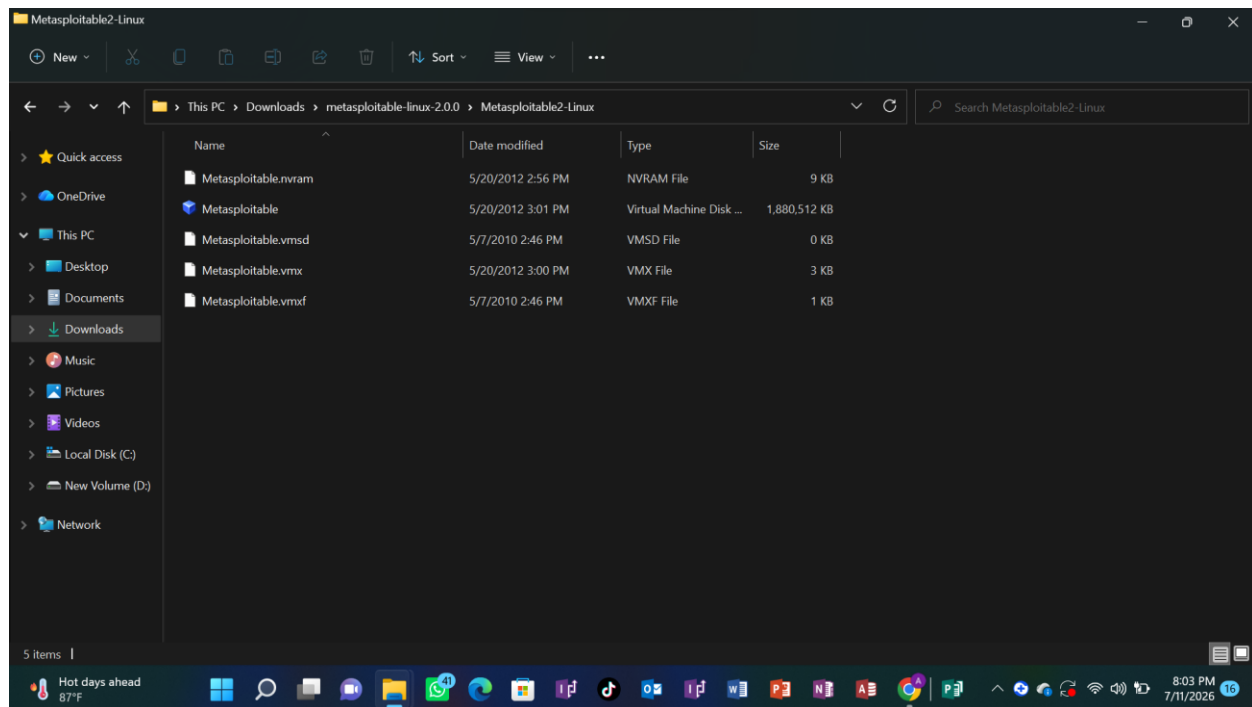
2. Extracting Metasploitable Files

After downloading the archive, it was extracted.

The extracted folder contains:

- Metasploitable.vmdk
- Metasploitable.vmx
- NVRAM file
- Other VMware files

These files are required for importing into VirtualBox.



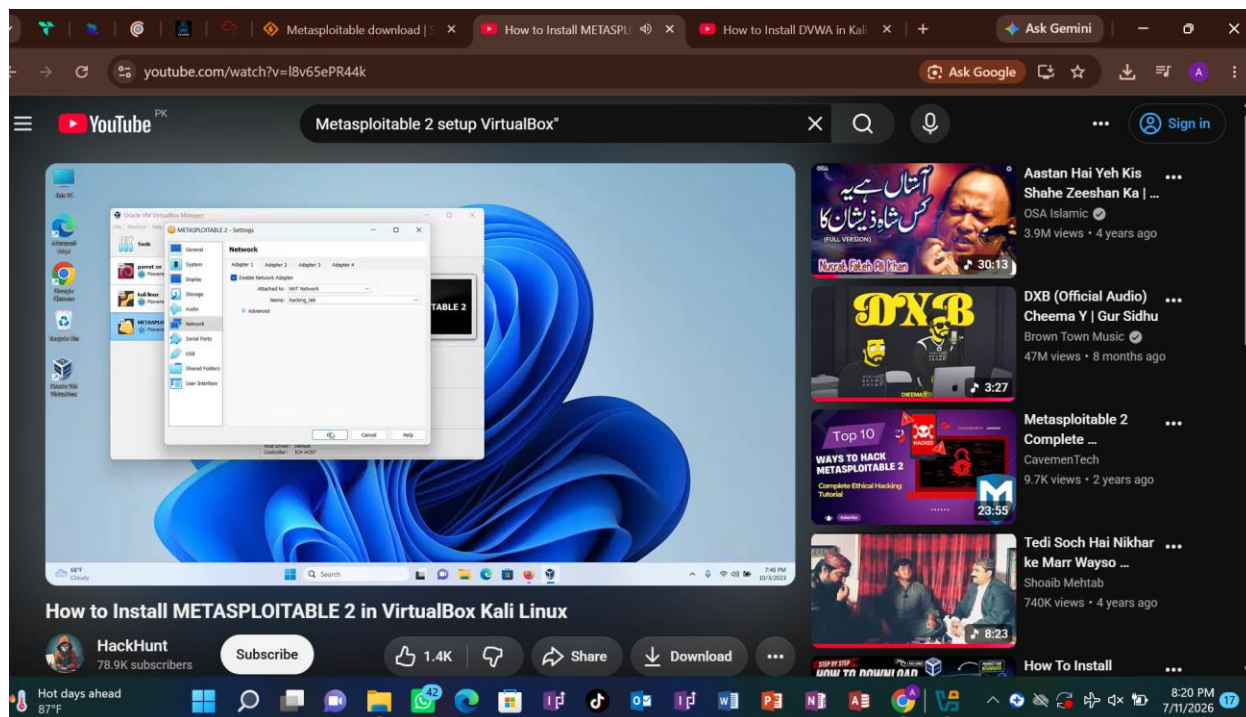
(Screenshot 2: Extracted Metasploitable Folder)

3. Following Installation Tutorial

A YouTube tutorial was followed to correctly configure the virtual machine in Oracle VirtualBox.

The tutorial demonstrated:

- Creating a new VM
- Selecting the existing virtual hard disk
- Network configuration



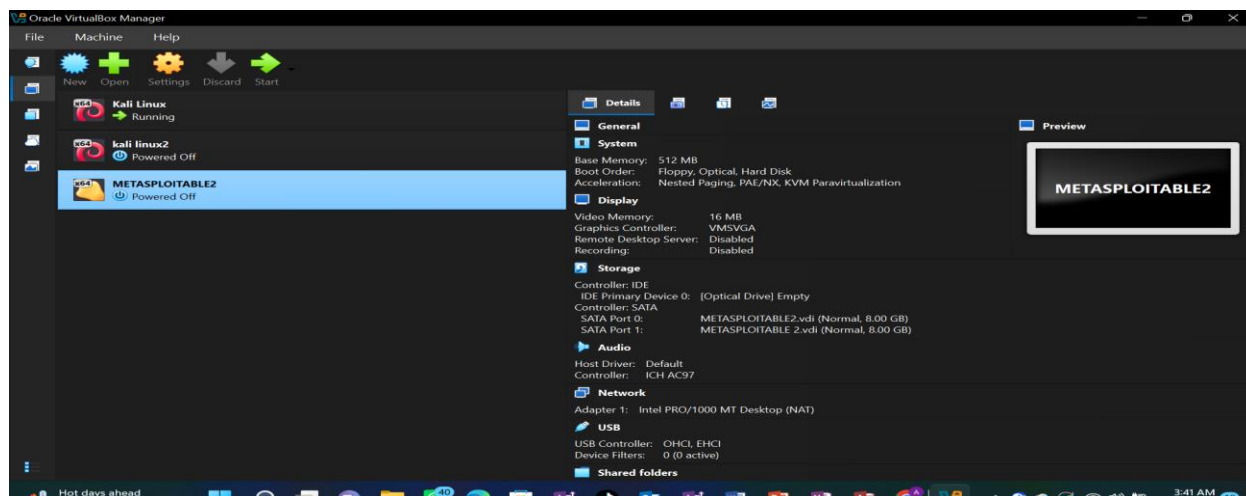
(Screenshot 3: YouTube Installation Tutorial)

4. Importing Metasploitable into VirtualBox

A new virtual machine named **Metasploitable2** was created inside Oracle VirtualBox.

The downloaded VMDK disk image was attached as the primary hard disk.

Memory allocation and hardware settings were configured according to the system requirements.



(Screenshot 4: Metasploitable VM Created in VirtualBox)

5. Starting Metasploitable

The virtual machine was successfully powered on.

Linux booted correctly and displayed the Metasploitable login screen.

Default credentials:

Username:

msfadmin

Password:

msfadmin

[illegible]

(Screenshot 5: Metasploitable Login Screen)

6. Finding the Target IP Address

After logging in, the following command was executed:

ifconfig

This displayed the assigned IP address of the target machine.

The target IP was noted because it will be used in scanning tasks later.

Target IP:

192.168.1.4



```
192.168.1.4: command not found

ali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.1.3  netmask 255.255.255.0  broadcast 192.168.1.255
    inet6 fe80::a80:27ff:fe3d:1804  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:3d:18:04  txqueuelen 1000  (Ethernet)
    RX packets 122  bytes 18044 (17.6 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 138  bytes 27670 (17.2 KiB)
    TX errors 0  dropped 2  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (local loopback)
    RX packets 8  bytes 480 (480.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 8  bytes 480 (480.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

ali@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3d:18:04 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.3/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 512sec preferred_lft 512sec
    inet6 fe80::a80:27ff:fe3d:1804/64 scope link noprefixroute
```

(Screenshot 10: Metasploitable ifconfig Output)

7. Configuring Kali Linux Network

Kali Linux network interface was verified using:

ifconfig

and

ip addr

These commands confirmed that Kali Linux had obtained an IP address on the same subnet.



(Screenshot : Kali Network Configuration)

8. Testing Initial Connectivity

Initially, Kali attempted to communicate using another network address.

The connection failed because both virtual machines were not yet configured on the same network.

The ping request returned errors.



(Screenshot 8: Initial Ping Failure)

9. Configuring Both Machines on NAT Network / Host-Only Network

Both virtual machines were configured to use the same VirtualBox network.

Network Adapter:

NAT Network

or

Host-Only Adapter

This allows direct communication between Kali Linux and Metasploitable.

```
* Starting deferred execution scheduler atd [ OK ]
* Starting periodic command scheduler crond [ OK ]
* Starting Tomcat servlet engine tomcat5.5 [ OK ]
* Starting web server apache2 [ OK ]
* Running local boot scripts (/etc/rc.local)
nohup: appending output to 'nohup.out'
nohup: appending output to 'nohup.out' [ OK ]

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: S_
```

(Screenshot 6: VirtualBox Network Configuration)

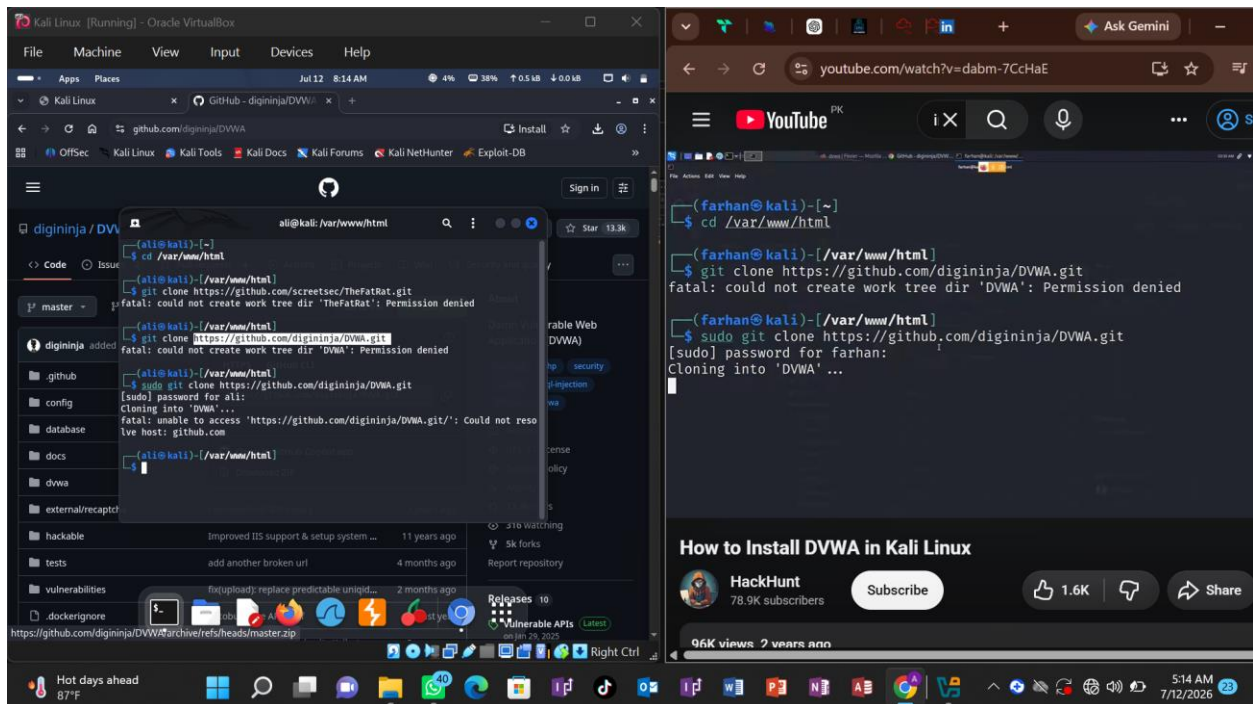
10. Verifying Connectivity

After correcting the network settings, Kali Linux successfully communicated with Metasploitable.

Command executed:

ping 192.168.1.4

The ping replies confirmed successful connectivity.



(Screenshot 12: Successful Ping)

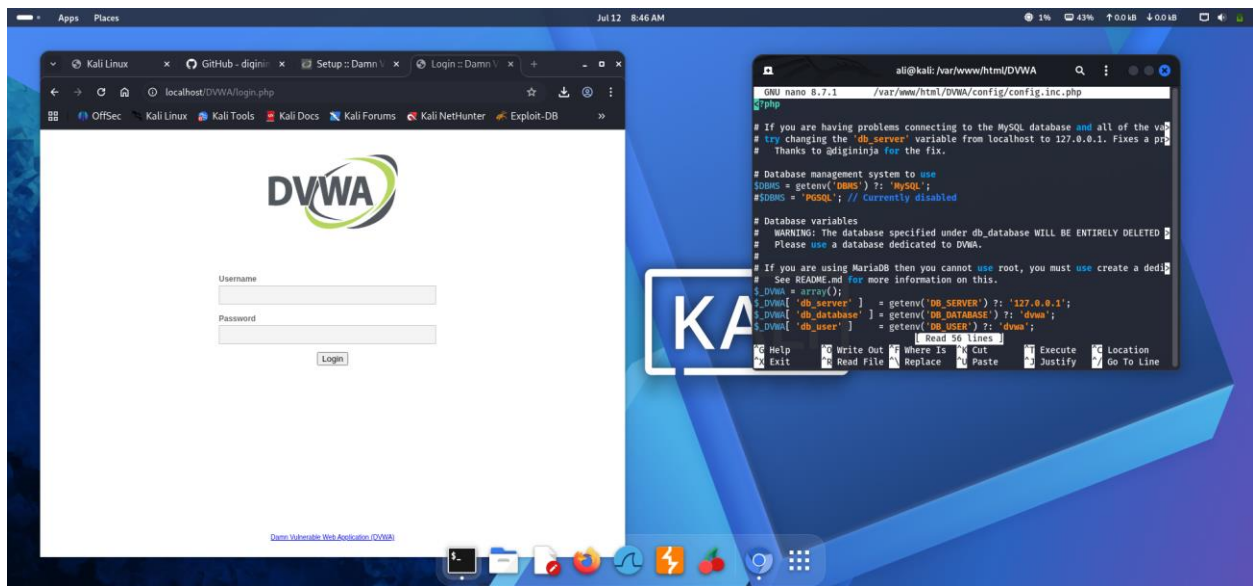
11. Installing DVWA

To practice web penetration testing, DVWA was installed inside Kali Linux.

The project repository was cloned from GitHub.

Initially, permission and DNS errors occurred.

These issues were resolved using sudo privileges and correct internet connectivity.



(Screenshot 13: Cloning DVWA Repository)

12. Configuring DVWA

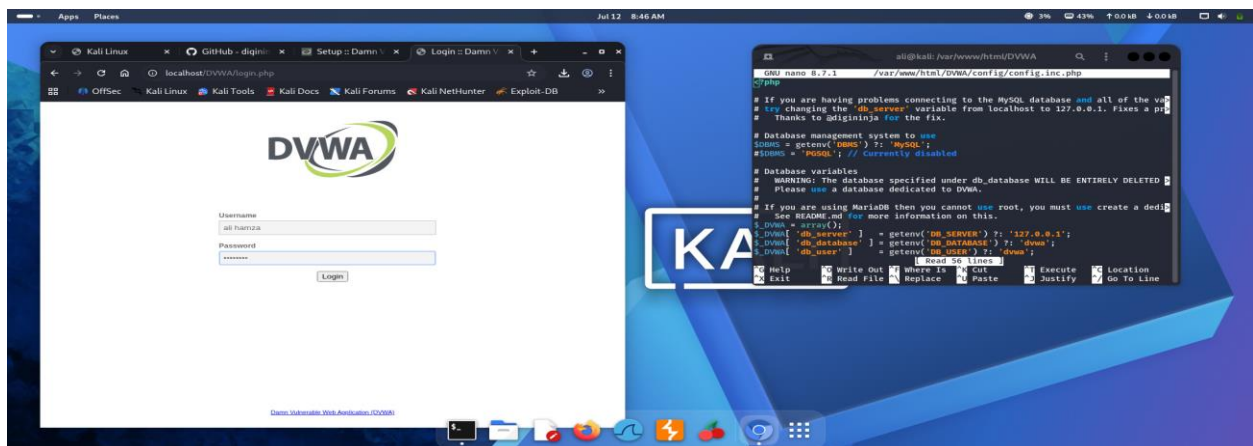
The configuration file

config.inc.php

was edited.

Database settings were modified according to the local MariaDB configuration.

The database username, password, and server details were updated.



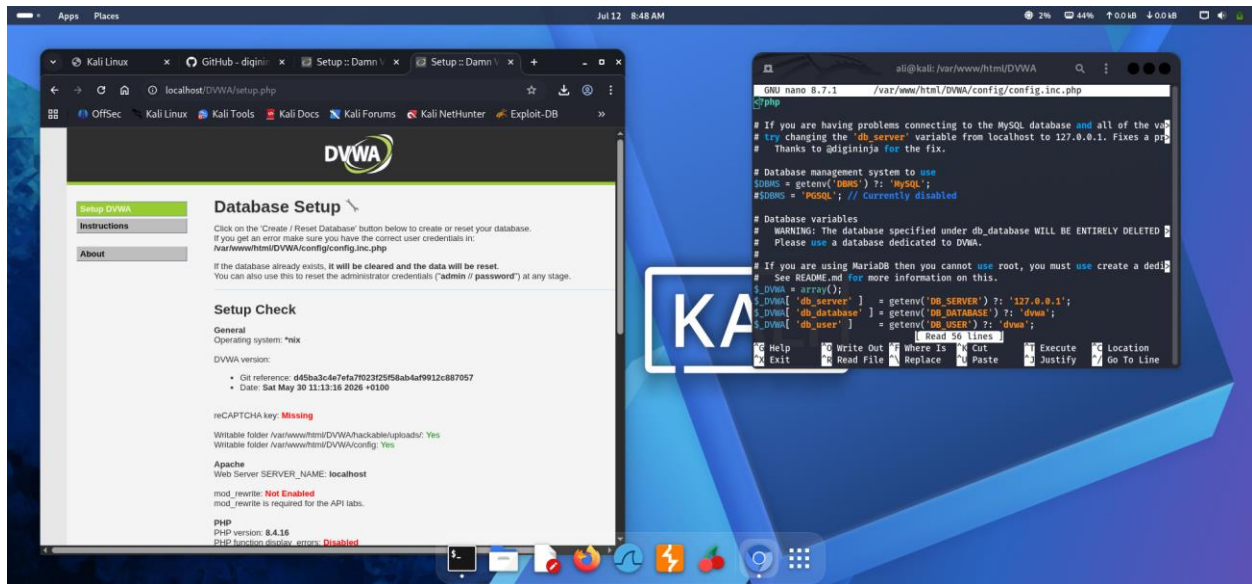
(Screenshot 14: Editing config.inc.php)

13. Accessing DVWA

DVWA was accessed through:

`http://localhost/DVWA`

The login page appeared successfully.



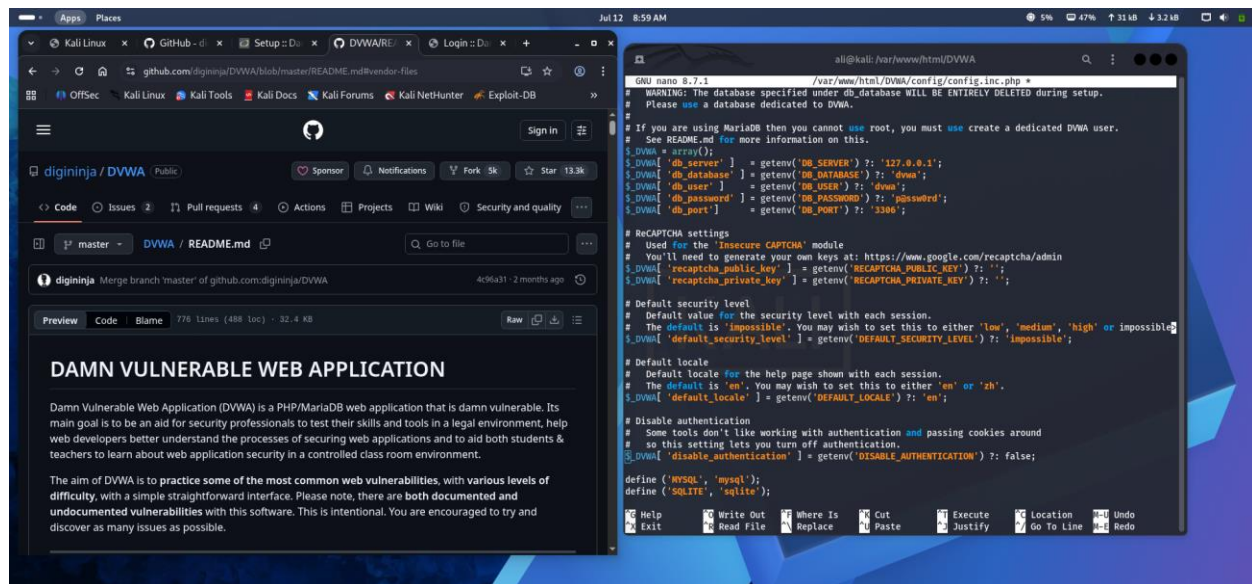
(Screenshot 15: DVWA Login Page)

14. Database Setup

The database setup page verified:

- PHP Installation
- Apache
- Database connectivity
- Writable folders

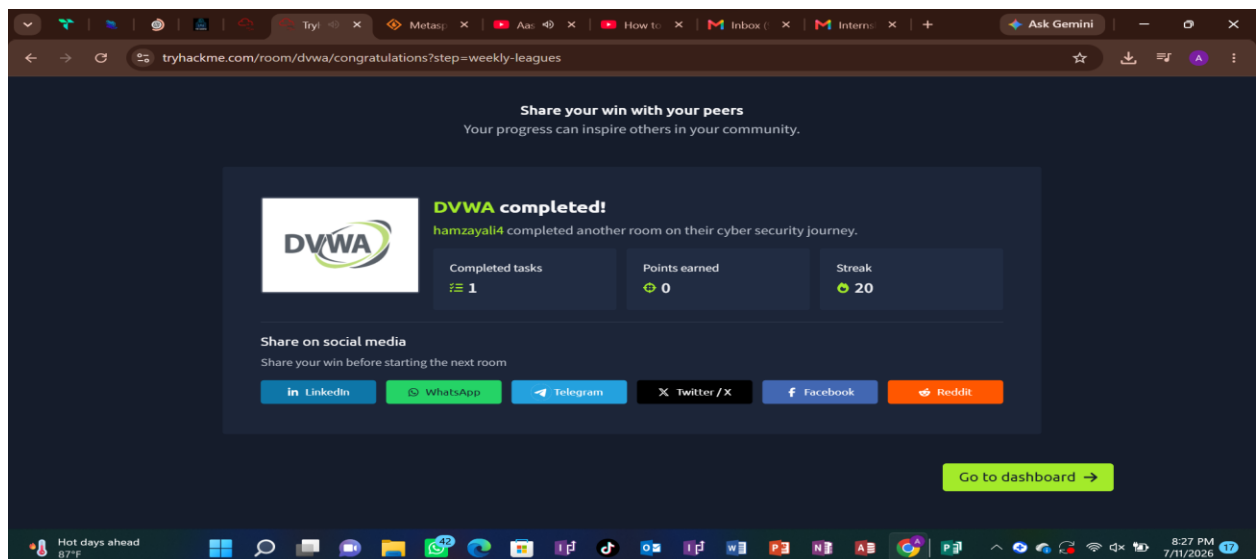
The "Create/Reset Database" option was used to initialize DVWA.



(DVWA Database Setup Page)

Resources Used

1. SourceForge – Metasploitable 2
2. GitHub – DVWA
3. Oracle VirtualBox
4. Kali Linux
5. YouTube – Metasploitable Installation Tutorial
6. YouTube – DVWA Installation Tutorial



7. TryHackMe – DVWA Practice Room

Conclusion

This lab successfully established a penetration testing environment using Oracle VirtualBox. Metasploitable 2 was downloaded, imported, and configured successfully. Kali Linux and the target machine were connected using the same virtual network, allowing successful communication through ICMP ping. DVWA was installed and configured on Kali Linux, providing a vulnerable web application for practicing web security testing. The environment is now fully prepared for future reconnaissance, vulnerability assessment, and exploitation exercises.

Learning Outcomes

After completing this lab, the following skills were achieved:

- Downloaded and imported Metasploitable 2 into Oracle VirtualBox.
- Configured virtual networking using NAT Network or Host-Only Adapter.
- Verified communication between Kali Linux and the target machine.
- Identified and recorded the target IP address for future scanning tasks.
- Installed and configured DVWA on Kali Linux.
- Prepared a complete penetration testing laboratory environment for upcoming cybersecurity exercises.

